

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-A)**DURATION: 25 MINUTES****SUMMER SEMESTER, 2024-2025****FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. Suppose the newly discovered “Hantavirus” affects 1% of the population. A test for this disease is 95% accurate. If a person tests positive, what is the probability they actually have the disease? 5
(CO1)
(PO1)

Solution:

Let D = has the disease and T = tests positive. $P(D) = 0.01$, $P(D') = 0.99$, $P(T|D) = 0.95$, $P(T|D') = 0.05$. Thus, $P(T) = P(T|D)P(D) + P(T|D')P(D') = 0.059$.

Using the Bayes' theorem, $P(D|T) = \frac{P(T|D)P(D)}{P(T)} = 0.161$

Rubric:

- 3 points for correctly finding the sub-terms.
- 2 points for correctly finding the solution.

2. If a loss function is non-convex and has multiple shallow local minimum values, should we use batch or stochastic gradient descent? 5
(CO1)
(PO1)

Solution:

SGD's noisy gradient estimates introduce random perturbations or noise at each step. If a local minimum is shallow (its basin is narrow and the loss advantage over nearby regions is small), the noise may be large enough to kick the solution out of it, allowing exploration of potentially better regions.

Rubric:

- 5 points for correct explanation.

3. Why is feature normalization important before applying gradient descent with L2 regularization? 5
(CO1)
(PO1)

Solution:

L2 regularization penalizes large weights equally. If features are on different scales, weights for small-scale features become artificially large just to compensate for the small feature values. The regularization term punishes these weights heavily, effectively penalizing that feature's importance unfairly.

Rubric:

- 5 points for correct explanation.